Worksheet 2.10

Getting into genes: Summary

Student: .... Class: ....

Use the listed words to complete the sentences below.

amino	chromosomes	diploid	dominant	errors	forensic
gametes	generation	genetics	genotypes	guanine	helix
identical	meiosis	nitrogen	opposite	physical	replicate
two					

1. The .... of cells contains genes made of DNA.
2. Genes determine many .... characteristics (e.g. eye colour).
3. Diploid human cells contain 23 chromosomal pairs. .... cells, however, contain only 23 chromosomes.
4. In a DNA molecule the number of thymine and adenine bases are always the same, and the amount of cytosine and .... bases are always the same.
5. The Watson–Crick DNA model is a double .... like a twisted ladder.
6. Each step of the DNA helix is made up of two .... bases and the bases are attached to the sugar units that are part of the sides of the ladder.
7. The steps of the DNA helix are composed of paired nitrogen bases. There is always an adenine base .... a thymine base and a cytosine base opposite a guanine base.
8. Polypeptides are made up of units called .... acids. A set of three DNA bases codes for a particular amino acid.
9. During cell division (mitosis) the DNA needs to .... to ensure the new cells have identical copies of the DNA.
10. In most instances the DNA is copied exactly during mitosis. In some cases .... occur and mutations result.
11. Gametes are formed by a type of cell division called ....
12. The fertilised egg has a .... number of chromosomes.
13. X and Y are the sex chromosomes. Females have .... X chromosomes and males have an X and a Y chromosome.
14. Different forms of a gene are called alleles. Alleles can be .... or recessive. Two copies of a recessive gene are required for the characteristic to be expressed.
15. Gregor Mendel was an Austrian monk who conducted the first quantitative .... experiments using garden peas.
16. Punnett squares are useful tools to predict the proportion of .... and phenotypes in the offspring.
17. Pedigree diagrams are useful in showing how various genes are transferred from one .... to another.
18. DNA fingerprinting is used in .... investigations and paternity tests.
19. Clones are organisms that are genetically ....